1 The kinetic energy E of an object rotating about its centre of mass is calculated using

$$E = \frac{1}{2} k\omega^2$$

where ω is the angular speed of rotation, and k is a constant.

What is the unit for k?

A kg m^2 **B** kg m s^{-1} **C** kg m s^{-2} **D** kg m^2 s^{-4}